

1. Administrative & Study Identification

Full Study Title: A Study of Subcutaneous Nivolumab Versus Intravenous Nivolumab in Participants With Previously Treated Clear Cell Renal Cell Carcinoma That is Advanced or Has Spread
Short Title/Acronym: CheckMate-67T (CA209-67T) **Protocol Number:** CA209-67T **NCT Number:** NCT04810078 **Posting ID:** 777995428491084814
Sponsor Name: Bristol-Myers Squibb **Investigational Product:** Subcutaneous Nivolumab (co-formulated with rHuPH20) vs. Intravenous Nivolumab **Phase of Development:** Phase 3 **Trial Status:** Active, Not Recruiting **Medical Monitor:** Bristol-Myers Squibb Medical Monitor

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the pharmacokinetics (PK) non-inferiority of subcutaneous (SC) nivolumab versus intravenous (IV) nivolumab, specifically measuring the time-average serum concentration over 28 days (Cavgd28) and the trough serum concentration at steady state (Cminss). **Secondary Objectives:** To evaluate the Objective Response Rate (ORR) by Blinded Independent Central Review (BICR) with a minimum of 6 months follow-up, additional PK parameters (Cmind28, Cmax1), Progression-Free Survival (PFS), Overall Survival (OS), and to assess the incidence of adverse events (AEs) and overall safety.

2.2 Background & Rationale To provide a more efficient treatment delivery method for patients with advanced or metastatic clear cell renal cell carcinoma (ccRCC). Subcutaneous nivolumab serves as an alternative to intravenous infusion, significantly reducing administration time (improving health care efficiency and patient experience) while lowering the risk of IV port-associated complications such as infections and phlebitis. An additional substudy evaluates drug level biocomparability of subcutaneous nivolumab manufactured using two different manufacturing processes.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Active Comparator (IV Nivolumab) **Blinding:** Open-label **Randomization:** 1:1 Randomized (Parallel Assignment)

3.2 Planned Enrollment & Duration Target **\$N\$:** 681 patients **Number of Sites:** Multicenter (Global) **Estimated Study Start:** March 2021 **Estimated Primary Completion:** October 2023

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Histological confirmation of renal cell carcinoma (RCC) with a clear cell component, including participants who may also have sarcomatoid features. 2. Advanced RCC (not amenable to curative surgery or radiation therapy) or metastatic RCC (Stage IV). 3. Measurable disease as defined by Response Evaluation Criteria in Solid Tumors (RECIST) v1.1 criteria within 28 days prior to randomization. 4. Received no more than 2 prior systemic treatment regimens, with intolerance or progression on or after the last treatment regimen received within 6 months prior to randomization. 5. Karnofsky Performance Status ≥ 70 at screening. 6. Must agree to follow specific methods of contraception, if applicable.

4.2 Exclusion Criteria 1. Untreated, symptomatic central nervous system (CNS) metastases. 2. Concurrent malignancy (present during screening) requiring treatment or history of prior malignancy active within 2 years prior to randomization. 3. Active, known, or suspected autoimmune disease. 4. Prior treatment with a programmed death receptor-1 (anti-PD-1), programmed death ligand-1 (anti-PD-L1), or cytotoxic T-lymphocyte-associated antigen-4 (anti-CTLA-4) antibody, or any other antibody or drug specifically targeting T-cell costimulation or checkpoint pathways. 5. Known human immunodeficiency virus (HIV) positive with an acquired immunodeficiency syndrome (AIDS) defining opportunistic infection within the last year, or a current CD4 count < 350 cells/ μ L (participants with HIV are eligible if on established ART for at least 4 weeks and monitored locally). 6. Serious or uncontrolled medical disorders including active severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) infection within approximately 4 weeks prior to screening. 7. Treatment with any live attenuated vaccine within 30 days of first study treatment.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: * **Experimental Arm (SC):** Subcutaneous nivolumab 1200 mg co-formulated with recombinant human hyaluronidase PH20 (rHuPH20) administered every 4 weeks. * **Comparator Arm (IV):** Intravenous nivolumab 3 mg/kg administered every 2 weeks. **Route of Administration:** Subcutaneous (SC) vs. Intravenous (IV). **Duration of Treatment:** Treatment continued until disease progression, unacceptable toxicity, withdrawal of consent, completion of 2 years of treatment, or death.

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care and established antiretroviral therapy (ART) for eligible HIV-positive participants. **Prohibited:** Treatment with any live attenuated vaccine within 30 days of first study treatment.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Tumor Assessment, Safety Labs
Baseline	Day 0	Randomization, First Dose Administration
Interim	Every 2 to 4 Weeks	Safety Labs, PK Sampling, Drug Administration, AE Monitoring
End of Study	Up to 2 Years	Final Vital Signs, Tumor Restaging, Survival Follow-up

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Pharmacokinetics (PK) non-inferiority, specifically time-average nivolumab serum concentration over 28 days (Cavgd28) and trough serum concentration at steady state (Cminss). **Measurement Method:** Central Laboratory Analysis of serum samples.

7.2 Secondary & Exploratory Endpoints * Objective Response Rate (ORR) by Blinded Independent Central Review (BICR) with a minimum of 6 months follow-up. * Trough serum concentration at day 28 (Cmind28). * Maximum serum concentration after the first dose (Cmax1). * Progression-Free Survival (PFS) and Overall Survival (OS). * Incidence of adverse events (AEs), serious adverse events (SAEs), and immunogenicity (anti-nivolumab antibodies).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard continuous collection of AEs, including local injection site reactions, laboratory abnormalities, and immune-mediated adverse events. **Serious Adverse Events (SAEs):** Captured according to protocol timelines and reviewed for causality. **Data Monitoring Committee (DMC):** Internal/External committee per sponsor standard operating protocols.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) for primary efficacy and pharmacokinetic outcomes; Safety Set for all AE/SAE evaluations. **Sample Size Justification:** Powered to detect PK non-inferiority boundaries for Cavgd28 and Cminss, as well as non-

inferiority in ORR. **Handling of Missing Data:** Censoring applied to time-to-event efficacy analyses (e.g., OS and PFS); standard descriptive statistics for missing laboratory markers.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Regular onsite and remote monitoring visits to evaluate protocol adherence and source data verification. **Audit**

Trail: Validated eCRF completion, automated edit checks, and data clarification forms monitored by the Sponsor.

11. Study Identifiers & Governance

Primary ID: CA209-67T **Posting ID:** 777995428491084814 **Registry Number:** NCT04810078 **Sponsor/Lead Organization:** Bristol-Myers Squibb **Study Title:** CheckMate-67T **Official Regulatory Title:** A Phase 3, Open-label, Randomized, Noninferiority Trial of Subcutaneous Formulation of Nivolumab Versus Intravenous Nivolumab in Participants With Advanced or Metastatic Clear Cell Renal Cell Carcinoma Who Have Received Prior Systemic Therapy

12. Clinical Framework (ccRCC Specific)

Primary Condition: Clear Cell Renal Cell Carcinoma (ccRCC)

Diagnosis Threshold: Advanced (unresectable) or Metastatic (Stage IV) disease progression after 1-2 systemic regimens **Medical**

Methodology: Anti-PD-1 checkpoint inhibitor therapy **Optimization**

Target: Pharmacokinetic and ORR non-inferiority of subcutaneous drug delivery

13. Study Procedures & Methodology

Management Pathway: Standard clinical pathway for advanced ccRCC

Education Protocol (HCP): Subcutaneous formulation (rHuPH20)

dispersion training, dose tracking, and immune-mediated AE

management **Education Protocol (Patient):** Education on treatment

schedules (Q2W vs Q4W) and anticipated local injection site

reactions **Quality Audit Mechanism:** Independent blinded

centralized review (BICR) for tumor progression and strict PK

serum tracking

14. Observation & Follow-Up Schedule

Total Duration: Up to 2 years of active treatment, followed by long-term survival follow-up **Onsite Visit Cadence:** Every 2 weeks

(for IV arm) or every 4 weeks (for SC arm) **Remote Monitoring Frequency:** For ongoing survival follow-up post-treatment
Checklist Review Interval: Per tumor assessment and RECIST v1.1 evaluation schedule

15. Sign-off & Approval

Role	Name	Signature	Date					
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]					
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]					
Statistician	[Name]	_____	[DD-MMM-YYYY]					